Chapter 17. Noninvasive Respiratory AIDS: Rocking Bed, Pneumobelt, and Glossopharyngeal Breathing

Nicholas S. Hill

Rocking Bed and Pneumobelt

The rocking bed and pneumobelt are noninvasive ventilators that were developed and saw their greatest use during the latter years of the polio epidemics but are used rarely today. They both rely on the effect of gravity to assist diaphragmatic motion and are particularly well suited to patients with severe diaphragmatic weakness or paralysis. Neither one should be used in the management of acute respiratory failure, and both have limited present-day applicability. Despite the similarities, there are also important differences, such as portability and suitability for nocturnal versus daytime use. This chapter reviews the historical development, mechanisms of action, and present-day uses of the rocking bed and pneumobelt. Glossopharyngeal breathing, another noninvasive approach to ventilator assistance, is discussed briefly at the end of the chapter.

Historical Development

The conceptual groundwork for development of the rocking bed was laid during the early 1930s by Eve,[1] who described the use of manual rocking to assist ventilation in two patients with acute respiratory paralysis. The technique consisted of placing the patient supine on a stretcher that was pivoted on a fulcrum placed at waist level. The patient then was rocked up and down approximately 45 degrees in either direction. Eve noted that the "weight of the viscera pushed the flaccid diaphragm alternatively up and down," achieving artificial respiration.[1] The technique was adopted subsequently by the British Navy as the recommended means of resuscitation for drowning victims.[2] Later studies demonstrated that this tilting method compared quite favorably with other resuscitation methods of the day, and it remained an acceptable means of resuscitation until mouth-to-mouth resuscitation gained acceptance during the 1960s.[3,4]

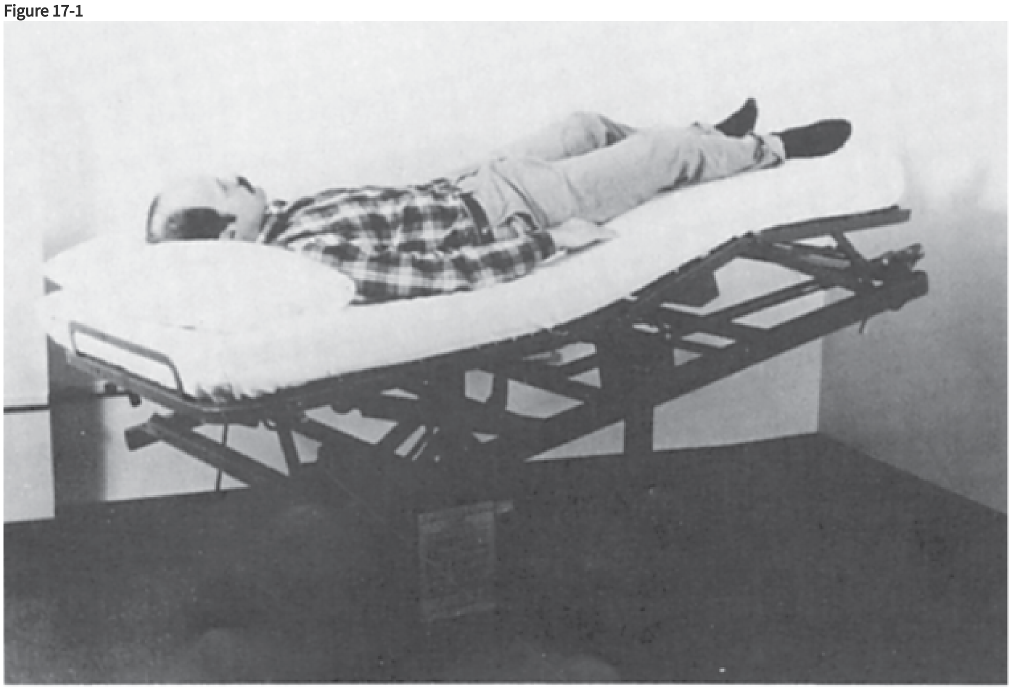
Automatic rocking beds were first introduced as ventilatory aids during the late 1940s. Wright[5] was the first to describe the management of respiratory insufficiency using an oscillating bed that had been designed originally to assist circulation. This experience led to the development of the McKesson Respiraid rocking bed, which was accepted by the Council on Physical Medicine and Rehabilitation in 1950.[6] Intended mainly as an aid to weaning patients with poliomyelitis from dependence on the tank respirator,[7] it facilitated nursing care and enhanced patient freedom but was quite noisy, bulky, and heavy (455 kg).[6,7] The Emerson rocking bed (J. H. Emerson Co., Cambridge, MA), also introduced during the late 1940s, was quieter and lighter than the McKesson bed and became the dominant model during the 1950s. Hundreds of rocking beds were manufactured between 1950 and 1960 (Emerson JH, personal communication), but after introduction of the Salk and Sabin vaccines and control of the polio epidemics, demand fell drastically. Many survivors of the polio epidemics continued to use rocking beds for ventilator support, sometimes for decades,[8] but most have since died or switched to other ventilators, and present-day use is rare.

The intermittent abdominal pressure respirator or insufflator (pneumobelt) was introduced at the end of the polio epidemics in an attempt to address the limitations of existing ventilators.[9] The pneumobelt was designed to allow complete freedom of the upper extremities and mouth during use in the sitting position and was intended mainly as a daytime ventilatory aid during meals or wheelchair use. Several modifications of the pneumobelt have been reported since the original description, but these never gained wide acceptance. These modifications include a piston-like device that compresses the abdomen while the patient sits in a wheelchair[10] and a combination of the pneumobelt and intermittent positive-pressure breathing.[11] Like the rocking bed, the pneumobelt has seen only limited use since control of the polio epidemics.

Mechanism of Action

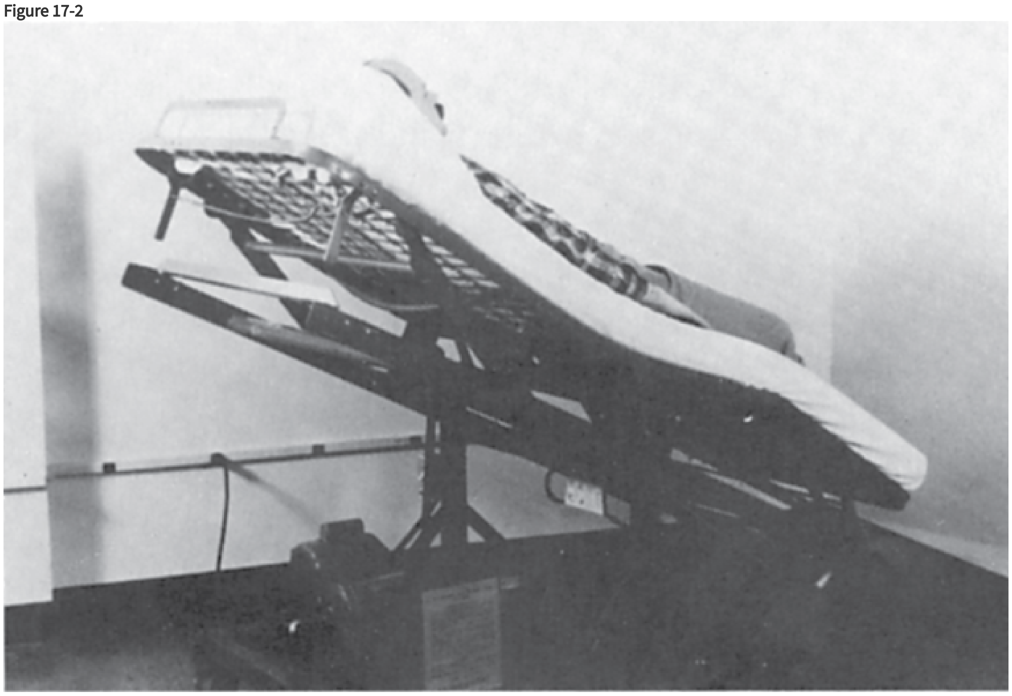
Eve compared the rocking-induced motion of the abdominal viscera within the thorax with that of a piston within a cylinder.[1] As the head moves down, the viscera and diaphragm slide cephalad, assisting exhalation (Fig. 17-1). In the foot-down position, the abdominal contents and diaphragm slide caudad, assisting inhalation (Fig. 17-2).

Figure 17-1



Rocking bed in 10-degree head-down position. Sliding of the viscera and diaphragm cephalad assists exhalation.

Figure 17-2

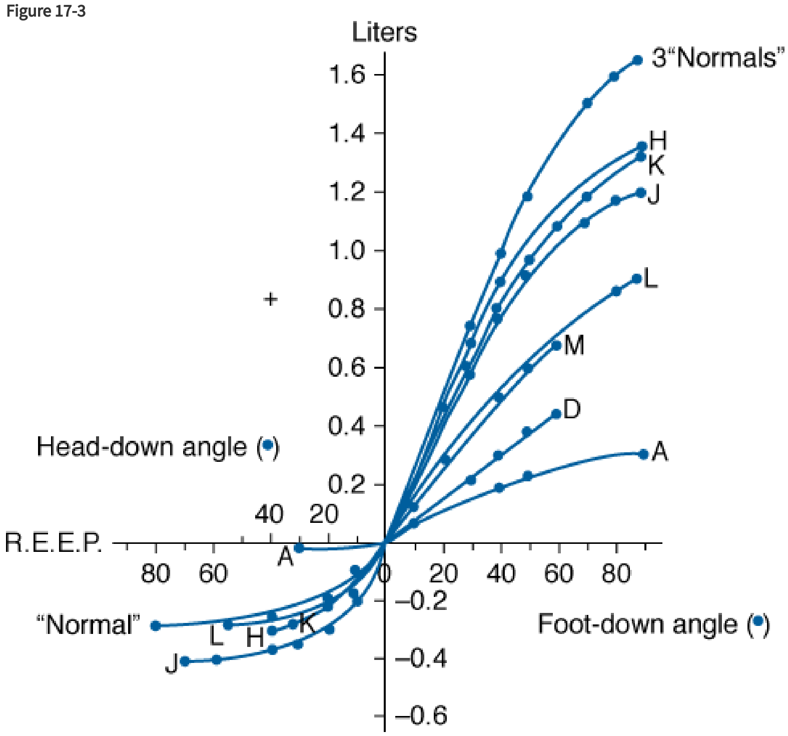


Rocking bed in 27-degree foot-down position. Sliding of the abdominal viscera and diaphragm caudad assists inhalation. (Used, with permission, from Hill.[17])

A number of early studies on the efficacy of rocking compared it as a method of resuscitation with others then used commonly.[3,4] These studies used fresh corpses or live subjects with pharmacologically induced paralysis or voluntarily suspended respirations to show that rocking produced tidal volumes ranging from a few hundred milliliters to a liter, and success in the resuscitation of near-drowning victims was reported.[2] Later, Plum and Whedon[12] found that the automatic rocking bed was not as effective as the tank respirator but produced adequate alveolar ventilation in eleven convalescent patients with respiratory paralysis secondary to poliomyelitis. The bed, however, was unable to sustain adequate ventilation in five patients during the acute stages of respiratory paralysis. Plum and Whedon recommended that use of the rocking bed be reserved for stable patients who are capable of at least some spontaneous breathing, a recommendation that holds true today.

Colville et al[13] subsequently examined the physiologic effects of rocking on respiratory mechanics and identified factors responsible for the wide individual variations in tidal volumes generated during rocking. They found that the greatest displacement of the diaphragm occurred during rocking from horizontal to the 40-degree foot-down position (Fig. 17-3). Beyond this angle, relatively little further displacement of the diaphragm occurred. Likewise, relatively little displacement of the diaphragm occurred during rocking from the horizontal to the head-down position. This indicated that in the horizontal position, the resting diaphragm was fairly close to its uppermost position, so the greatest passive motion could be achieved by applying gravitational force in the caudad direction.

Figure 17-3



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Relationship between angle of rocking bed and static lung volume in three normal individuals and in patients with poliomyelitis, as indicated by letters. Note that the greatest volume shift occurs between 0 degrees and the 40-degree foot-down angle. Note also the marked variability between individuals. *REEP,* resting end-expiratory position. (Used, with permission, from Colville P, et al. Effects of body tilting on respiratory mechanics. *J Appl Physiol.* 1956;9:19–24.)

Along these lines, Joos et al[14] found that tidal volumes were greater in some patients when rocking was achieved entirely in the head-up position (5 to 42 degrees above the horizontal plane) rather than between the head-down and head-up positions (10 degrees below to 27 degrees above the horizontal plane). In addition, others found that relatively little increase in minute volume could be achieved by increasing the rate of rocking beyond 15 to 16 rocks per minute.[3] At higher rocking frequencies, the tidal volume tended to diminish, negating the effects of the increased rocking rate.

With regard to the large individual variation (see Fig. 17-3), Colville et al[13] found that as the compliance of the abdominal wall increased, the greater was the tidal volume during rocking. Thus, rocking was relatively ineffective in patients with severe kyphoscoliosis, who have low abdominal and diaphragmatic compliance and short abdominal lengths. Taken together, these findings indicate that the efficacy of the rocking bed is highly dependent on patient body characteristics and that function is likely to be optimal when rocking is between the near-horizontal and the 40-degree foot-down positions at rocking rates between 12 and 16 rocks per minute. Wider manipulations of rocking rate and arc, however, may be useful in optimizing alveolar ventilation in some patients.

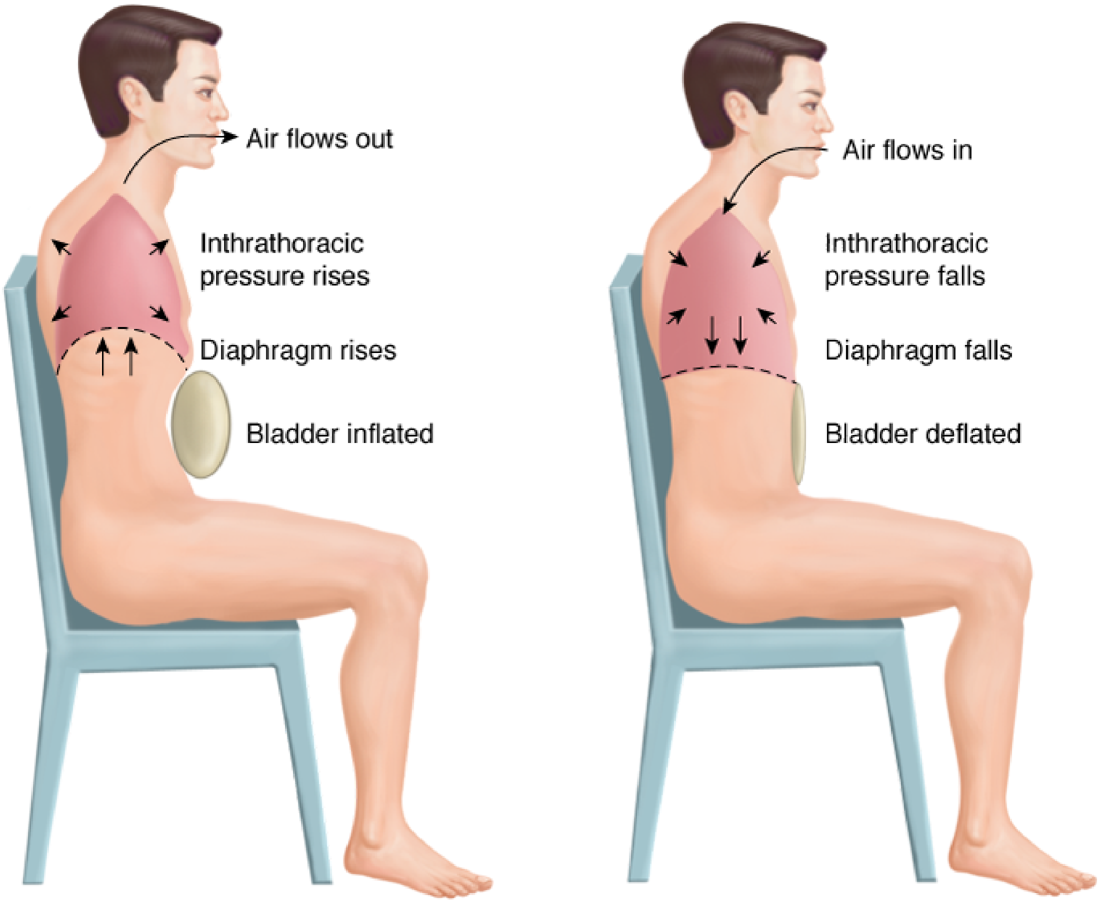
The pneumobelt[9] operates by a mechanism similar to that of the rocking bed in that it assists diaphragmatic motion by causing piston-like motions of the abdominal viscera within the thoracic "cylinder." The major difference is that the pneumobelt assists exhalation by applying positive pressure to the abdominal surface rather than using gravitational force. It consists of an inflatable rubber bladder held firmly over the abdomen by an adjustable corset (Fig. 17-4). Inflation of the bladder squeezes the viscera, forcing the diaphragm cephalad and actively assisting exhalation. On deflation of the bladder, gravity pulls the viscera and diaphragm back to their original positions, assisting inhalation (Fig. 17-5). Because of this dependence on gravitational force, the pneumobelt must be used in a sitting position, optimally at angles of 45 degrees or greater. Below 30 degrees, the ability to assist ventilation is diminished markedly. Thus, nocturnal use of the pneumobelt is limited to patients who can sleep in a sitting position.[15]

Figure 17-4

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Pneumobelt is shown attached via a connecting hose to a Bantam positive-pressure ventilator (Puritan-Bennett Corp, Lenexa, KS). (Used, with permission, from Rondinelli et al. In: Delisa JA, ed. *Rehabilitation Medicine: Principles and Practice.* Philadelphia, PA: Lippincott; 1988.)

Figure 17-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

The pneumobelt functions by exerting positive pressure on the abdominal viscera, forcing the diaphragm up and assisting exhalation (*left*). When the bladder deflates, gravity returns the diaphragm to its original position, assisting inhalation (*right*). (Reproduced with the permission from the American College of Chest Physicians. Hill NS. Clinical applications of body ventilators. *Chest.* 1986;90:897–905.)

Despite this limitation, the pneumobelt can be a very useful device in appropriately selected patients. The ability of the pneumobelt to augment tidal volume is linearly related to the inflation pressure of the bladder between pressures of approximately 15 and 50 cm H_2O. Pressures exceeding 50 cm H_2O are rarely tolerated because of abdominal discomfort. As with the rocking bed, the ability of the device to augment tidal volume varies considerably among individuals. Important factors that contribute to the variability among individuals include abdominal and chest wall compliances. High abdominal compliance favors efficient functioning of the device, but high chest wall compliance may allow expansion of the chest wall during bladder inflation, reducing efficiency.[9] Thus, like the rocking bed, body habitus is also an important determinant of ventilator efficiency for the pneumobelt. Although thin patients can be ventilated effectively,[16] both devices work less efficiently in extremely thin or obese patients and in those with severe kyphoscoliosis. The sitting position in patients with severe kyphoscoliosis often brings the lower rib cage in contact with the thighs, rendering proper positioning of the pneumobelt impossible.

Application

Rocking beds are no longer available commercially. Pneumobelts (small, medium, or large) can be obtained via home medical equipment vendors via Philips Respironics, Inc. (Murrysville, PA). Positive-pressure volume-limited ventilators with sufficient pressure-generating and volume-generating capabilities can be used. Most portable “bilevel” positive-pressure devices, however, are insufficient.

Consisting of a bed frame that is suspended on an axis 100 cm above the ground, rocking beds are bulky (193 cm long by 84 cm wide) and heavy (approximately 136 kg). Although the axis of rotation can be adjusted, they usually are set to rotate 10 degrees in the head-down direction and 27 degrees foot-down. As noted previously, maintaining the rocking arc in the head-up range may improve efficacy in some patients.[14] Adjustable cranks allow raising of the head and knees to minimize slippage. Excessive flexion of the hips, however, may impair functioning.[14] A foot rest also may be attached to prevent downward sliding, but with proper positioning of the head and knees, this is usually unnecessary.

The most attractive feature of the rocking bed is its ease of application. With use of a foot stool, most patients can position themselves on it with minimal assistance. Paralyzed patients require help from one or two attendants depending on body weight. The patient lies supine with the head and knees raised slightly to maximize comfort. A baseline minute volume is measured using a portable spirometer. Rocking then is commenced at rates between 12 and 16 rocks per minute, again adjusted to optimize patient comfort. The patient is coached to exhale during head-down rocking and to inhale while the head moves up. When synchronization has been achieved, further adjustments in rocking rate may be made to optimize minute volume. An arterial blood-gas determination may be helpful after an hour or two during the initial trial to assess adequacy of alveolar ventilation.

Patients then are encouraged to initiate rocking at bedtime and to extend the hours of use gradually until they are able to sleep through the night. Most patients find the rocking bed quite comfortable and encounter little difficulty in learning to sleep while rocking. Motion sickness is unusual, presumably because the bed rocks in one plane only, but it may disturb some patients. The bed limits daytime mobility, so most patients use it only nocturnally. If daytime ventilatory assistance is necessary, another technique, such as noninvasive positive pressure ventilation or the pneumobelt, is recommended for daytime supplementation.

Application of the pneumobelt is slightly more difficult than that of the rocking bed, usually requiring at least one attendant unless the patient has full upper extremity strength. The corset is positioned while the patient is sitting comfortably. The rubber bladder is positioned over the abdomen with the curved lower border of the corset over the pubis and the horizontal upper border over the xiphoid. Some coverage of the lower rib cage may serve to minimize paradoxical motion of the ribs during assisted exhalation, although some authors have found more efficient functioning if the corset is placed below the xiphoid.[16] If the corset extends too far below or above the lower rib margin, another of the three available sizes should be tried. The corset then is tightened using the three straps that surround the abdomen until the rubber bladder is held firmly but not uncomfortably against the abdomen. After baseline spontaneous respiratory rate and tidal volume are measured, the rubber bladder is attached to a positive-pressure ventilator using a connecting hose. Volume-limited positive-pressure ventilators that generate adequate pressures and volumes will operate the pneumobelt successfully, but portable “bilevel-type” pressure-limited devices that are popular to provide noninvasive positive-pressure ventilation are not adequate because their pressure-generating capabilities are too limited.

The ventilator is set at a rate that approximates spontaneous respiratory frequency and an inspiratory-to-expiratory (I:E) ratio of approximately 1:1.5. Some authors have found that function is optimal at rates of 12 to 14 breaths/min,[16] but this may depend on the underlying respiratory disorder. Some patients prefer rates as high as the low 20s per minute (see “Case 2: Use of the Pneumobelt” below). Ventilator assistance is initiated by intermittently inflating the rubber bladder with peak inflation pressures of 20 to 25 cm H_2O. Inflation pressure then is raised gradually until the patient’s assisted tidal volume increases to the desired range, usually 30% to 50% over spontaneous breathing or the patient reaches the limit of tolerance. Peak pressures of 30 to 50 cm H_2O are usually sufficient, but pressures up to 60 cm H_2O may be necessary in some patients.[16] Pressures exceeding 60 cm H_2O rarely are tolerated because of discomfort. Considerable coaching usually is necessary during initial adaptation to encourage synchronization of patient breathing with ventilator cycling. Patients who will be successful usually learn to synchronize their breathing with the ventilator after a few sessions.

In addition to measurement of minute volume, efficacy is assessed using arterial blood-gas determinations that are done as soon as the patient is comfortable and synchronizing well. The desired amount of ventilator assistance will vary depending on the patient, but a decrease in arterial carbon dioxide tension (Pa_{CO_2}) of approximately 5 to 10 mm Hg below spontaneous breathing levels is acceptable during the initial sessions. Subsequent use of the device also depends on the patient. Most often, because the pneumobelt must be used in the sitting position, it is used as a daytime ventilator aid to supplement another device used nocturnally. As illustrated by Case 2 (see below) and another case report,[15] however, occasional patients may adapt to nocturnal use. The monthly rental for the belt itself is less than $100, but ventilator costs and associated respiratory therapy services raise monthly charges to the $500 to $1000 range.

Long-term follow-up of patients using either the rocking bed or pneumobelt should include assessment for symptoms or signs of chronic hypoventilation such as morning headache, hypersomnolence, and ankle swelling. Daytime spontaneous arterial blood-gas determinations are particularly useful for assessing the adequacy of ventilatory assistance,[17,18] and attempts should be made to increase assisted minute volume if Pa_{CO_2} remains substantially above 50 mm Hg. Nocturnal oximetry or polysomnography are useful not only to assess adequacy of nocturnal oxygenation but also to detect obstructive apneas that may be induced by negative-pressure ventilators[19] and also may occur during use of the rocking bed or pneumobelt.

Complications of rocking bed or pneumobelt use are relatively few. Some patients using the pneumobelt for many consecutive hours develop skin abrasions, and others have trouble coordinating their breathing with either of the ventilators so that efficacy may be inadequate. If appropriate patients are selected, the risk of worsened respiratory failure or arrest during use is low because such patients should be capable of some spontaneous breathing. As deterioration occurs with progressive neuromuscular syndromes or acute respiratory infections, however, patients may have to switch to other, more effective ventilators.

Indications

Considering that the rocking bed and pneumobelt share similar mechanisms of action, it is not surprising that they also share indications for use (Table 17-1). Because both assist ventilation essentially by augmenting diaphragmatic motion, they are particularly useful in patients with bilateral diaphragmatic weakness or paralysis. Abd et al[20] demonstrated the utility of this application in patients with bilateral diaphragmatic paralysis following open-heart surgery. In this study, ten patients were weaned from conventional positive-pressure ventilation to the rocking bed and continued to use it nocturnally until phrenic nerve function recovered after 4 to 27 months.

Table 17-1: Indications and Contraindications for the Rocking Bed and Pneumobelt

Indications
Chronic respiratory failure[a] caused by:
Bilateral diaphragmatic paralysis
Muscular dystrophies
Duchenne
Limb-Girdle
Myotonic
Postpolio syndrome
Amyotrophic lateral sclerosis
Multiple sclerosis
Traumatic quadriplegia[b]
Contraindications
Acute respiratory failure or rapidly progressive neuromuscular disease
Excessive secretions
Upper airway dysfunction
Excessive obesity or thinness
Severe kyphoscoliosis

[a] With intact upper airway and appropriate body habitus.

[b] Pneumobelt only.

The rocking bed and pneumobelt also may be used in the management of chronic respiratory failure caused by a variety of slowly progressive neuromuscular syndromes that weaken the diaphragm and leave upper airway function intact. Chalmers et al[21] described their experience with fifty-three neuromuscular patients, thirty with postpolio syndrome, twelve with various muscular dystrophies, four with adult-onset acid maltase deficiency, and four with motoneuron disease (amyotrophic lateral sclerosis). Forty-three of the patients used the rocking bed for an average of 16 years with good control of symptoms and stabilization of gas exchange in most patients. Seventeen patients discontinued use: nine because of discomfort and eight because progression of respiratory insufficiency necessitated more efficacious therapy. The pneumobelt has been used in patients with high spinal cord lesions, mainly as a daytime supplement, often in combination with positive pressure ventilation administered noninvasively or via a tracheostomy.[16] In this setting, the pneumobelt frees the face of encumbrances and improves speech and mobility. It also may be used for total ventilator support.[22] Surprisingly, those with the least ability to sustain spontaneous breathing adapt best to pneumobelt use.[22]

Because both ventilators have limitations that are influenced heavily by the patient's body habitus, however, a number of caveats should be borne in mind (see Table 17-1). Neither device is suitable for use in acute respiratory failure mainly because a period of adaptation is necessary for optimal efficiency and also because excessive secretions interfere with function.[12] In addition, care should be exercised to select patients who have an appropriate body habitus, adequate upper airway function, and a condition that is sufficiently stable or slowly progressive so that the anticipated duration of use will justify the effort and time required for adaptation.

Because the rocking bed is suited for nocturnal use and the pneumobelt for daytime use, the two may be used in complementary fashion. A patient might use the rocking bed during sleep and the pneumobelt for daytime desk work or wheelchair use.

Many choices are available currently for noninvasive ventilation of patients with chronic respiratory failure secondary to idiopathic diaphragmatic paralysis or slowly progressive neuromuscular conditions. Noninvasive positive-pressure ventilation administered via a nasal mask is unquestionably the mode of first choice because of its convenience, ease of application, portability, and avoidance of the intermittent upper airway obstructions and oxygen desaturations associated with the use of negative-pressure ventilators.[19,23] Nevertheless, other noninvasive ventilators still should be considered in patients with chronic respiratory failure who are unable to use noninvasive positive-pressure ventilation, as illustrated in a recent report of a woman with scapuloperoneal muscular dystrophy who used a rocking bed successfully when loss of upper extremity strength rendered her incapable of applying her nasal ventilator.[24]

Efficacy comparisons between the rocking bed and pneumobelt and other noninvasive ventilators are limited because few relevant studies are available. In acutely anesthetized intubated subjects, Bryce-Smith and Davis[25] showed that the rocking bed produced barely adequate tidal volumes when rocking through an arc of 40 degrees and was much less effective than negative-pressure ventilators, including the tank ventilator and chest cuirass. Goldstein et al[26] demonstrated that cuirass-type negative-pressure ventilators augment tidal volumes and suppress diaphragmatic electromyographic activity more effectively than did the rocking bed in patients with neuromuscular disease. Both types of ventilators, however, were deemed effective, and considering that this was an acute daytime study, the rocking bed may have been more effective at reducing diaphragmatic electrical activity had patients been monitored overnight after a suitable adaptation period. Even so, the conclusion that the rocking bed is less effective than negative-pressure ventilators seems justified. In one case, the rocking bed was combined with noninvasive positive-pressure ventilation via a nasal mask to enhance efficacy and achieve "necessary ventilatory support."[27]

In summary, both the pneumobelt and rocking bed currently have limited indications (see Table 17-1). They are unsuitable for use in acute respiratory failure, must be used in patients with a relatively "normal" body habitus, and have marginal efficacy even under optimal circumstances. Noninvasive positive-pressure ventilation has convenience and efficacy advantages over these devices. Nonetheless, there are occasional patients who are unable to tolerate a nasal mask or negative-pressure ventilator who may prefer the relatively greater comfort of the rocking bed or the daytime freedom that the pneumobelt affords. The following cases illustrate such applications.

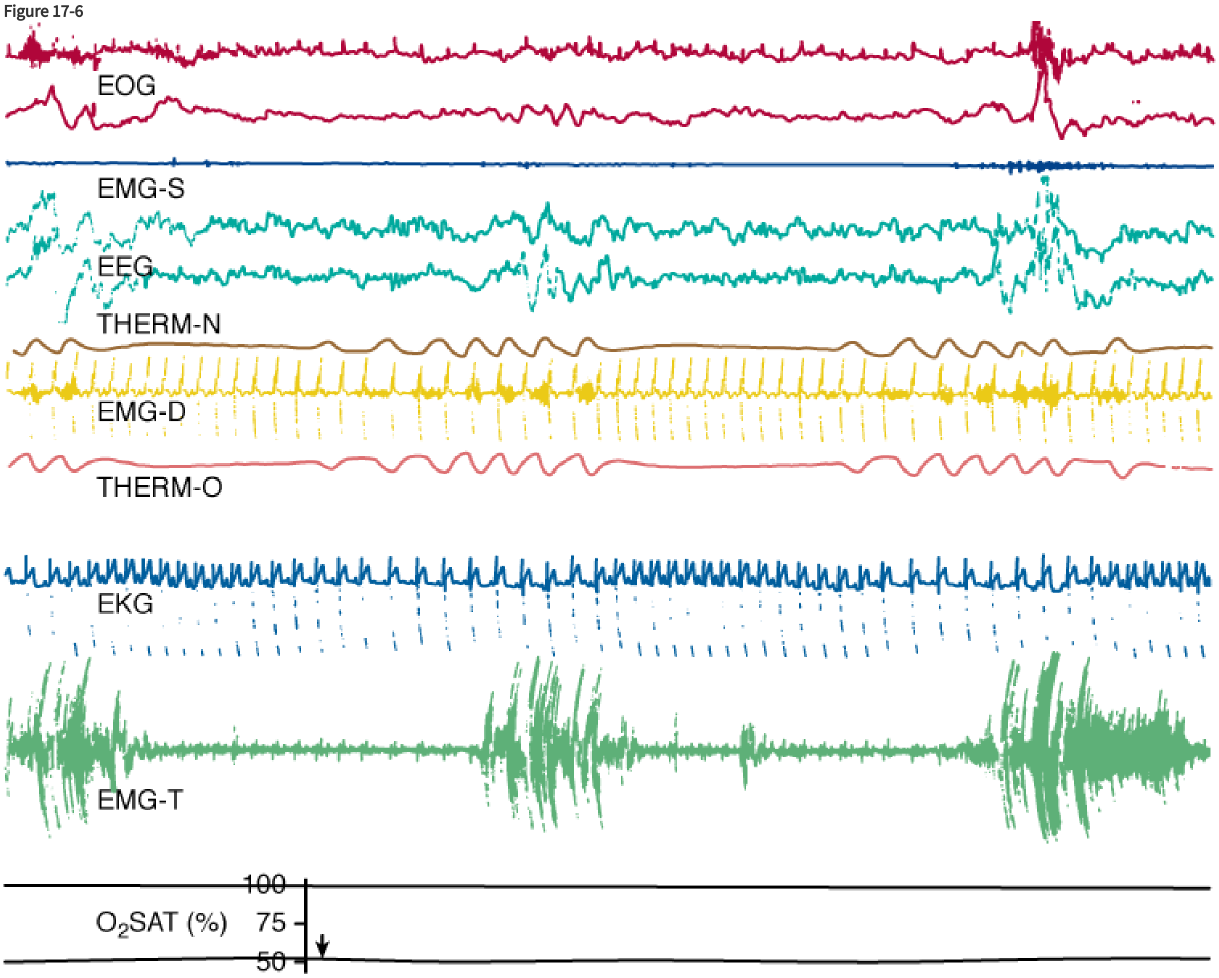
Case 1: Use of the Rocking Bed

J.F., a 47-year-old man with myotonic dystrophy, first presented with symptoms of chronic hypoventilation, including morning headache and daytime hypersomnolence. He was found to have global muscular weakness, although he was still able to walk with a cane. Pulmonary function studies demonstrated a severe restrictive defect with a forced vital capacity (FVC) of 1.5 L (31% of predicted) and a forced expiratory volume in 1 second (FEV_1) of 1.3 L. Room-air arterial blood-gas determinations revealed a pH of 7.36, a Pa_{CO_2} of

53 mm Hg, and a Pa_{O_2} of 69 mm Hg. Nocturnal polysomnography showed mild obstructive sleep apnea and sustained mild oxygen desaturations consistent with hypoventilation. A trial of noninvasive positive-pressure ventilation via a nasal mask was initiated. Despite much coaching and trials with different masks, the patient declined further use after 2 months because of intolerable mask discomfort. Megestrol, 40 mg PO TID, then was begun, followed within a month by normalization of blood gases and resolution of symptoms.

Two years later, the patient again developed symptoms of morning headache and daytime hypersomnolence. A daytime arterial blood-gas determination showed a pH of 7.36, a Pa_{CO_2} of 65 mm Hg, and a Pa_{O_2} of 42 mm Hg. Nocturnal polysomnography was repeated (see Fig. 17-6), showing periodic breathing and sustained severe oxygen desaturations. Oxygen supplementation was begun, but the Pa_{CO_2} rose to 76 mm Hg. Nasal ventilation was tried once more and was rejected rapidly by the patient. Initiation of negative-pressure ventilation using a pneumowrap brought about a rapid improvement in symptoms and daytime gas exchange, with a room-air arterial blood-gas determination showing a pH of 7.42, a Pa_{CO_2} of 55 mm Hg, and a Pa_{O_2} of 65 mm Hg. The patient's wife, however, refused to consider use of the pneumowrap at home because placing the patient in it each night seemed too difficult.

Figure 17-6

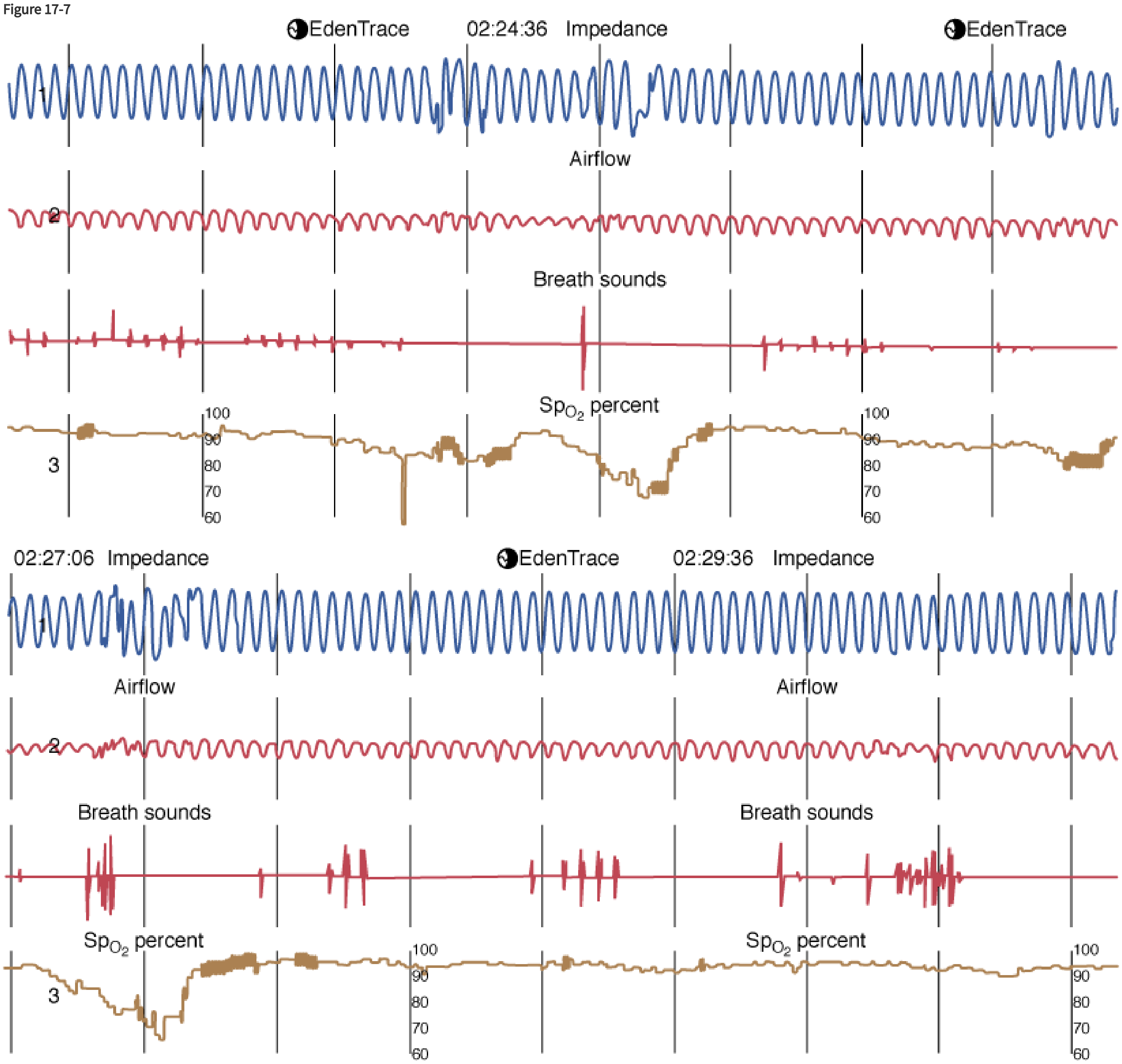


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Nocturnal polysomnogram in patient 1, obtained in August 1991 during spontaneous room-air breathing, shows periodic breathing with sustained severe oxygen desaturation. *EEG,* electroencephalogram; *EKG,* electrocardiogram; *EMG,* electromyelogram; *EMG-D,* diaphragmatic EMG; *EMG-S,* submental electromyogram; *EMG-T,* temporal EMG; *EOG,* electrooculogram; *l,* left; *O_2SAT,* oxygen saturation; *r,* right; *THERM-N,* nasal thermistor; *THERM-O,* oral thermistor.

Nocturnal ventilation using the rocking bed then was begun at a rocking rate of 16 rocks per minute. Tidal volume during the initial rocking trial ranged from 300 to 350 mL. The patient's wife found the rocking bed more acceptable because of simpler application. After a several-day adaptation period, the patient was able to sleep for 4 to 5 hours using the device, and he was discharged home. Nocturnal polysomnography obtained after several months of use showed good synchronization of chest wall motion with the rocking bed, but some obstructive hypopneas and apneas continued to occur (Fig. 17-7). The oxygen desaturations associated with the obstructive events were ameliorated by 2 liters/min of nasal oxygen used nocturnally (Fig. 17-8). The patient slept 7 to 8 hours nightly using the rocking bed with supplemental oxygen, and a daytime unassisted room-air arterial blood-gas determination after 15 months showed a pH of 7.42, a Pa_{CO_2} of 52 mm Hg, and a Pa_{O_2} of 78 mm Hg. After 3 years of clinical stability with nocturnal rocking bed use, the patient was hospitalized with pneumonia and progressive respiratory failure; he declined intubation and died.

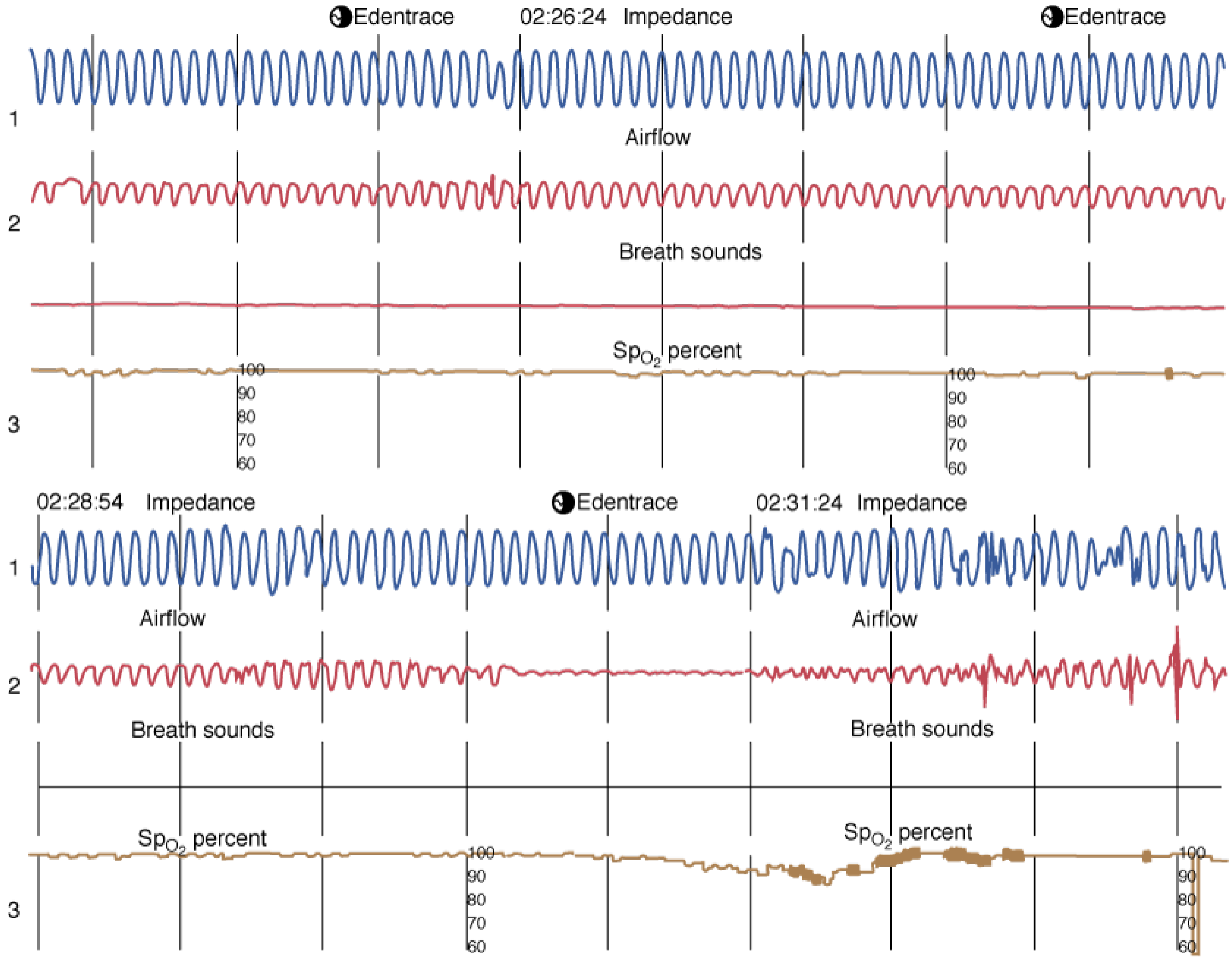
Figure 17-7



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Continuous nocturnal recording of case 1 during rocking bed use without oxygen supplementation. Tracing shows consistent regular chest wall motions at a rate of 16 breaths/min, indicating good synchronization with the rocking bed. However, periodic obstructive hypopneas are signified by decreases in airflow followed by oxygen desaturations and arousals, as evidenced by disruption of chest wall synchrony. *Channel 1* is chest wall impedance, *channel 2* is combined nasal and oral thermistor, and *channel 3* is finger pulse oximetry. *Vertical lines* indicate 30-second intervals. Monitoring was performed using Edentec monitor (Edentec, Inc., Eden Prairie, MN).

Figure 17-8



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Nocturnal recording of case 1 using rocking bed and 2 L/min O_2 via nasal cannula. Tracing again shows excellent synchrony of chest wall motion with rocking. A 50-second obstructive apnea is apparent in the lower tracing, but is associated with only a mild oxygen desaturation to 90%. *Channel 1* is chest wall impedance, *channel 2* is combined nasal and oral thermistor, and *channel 3* is finger pulse oximetry.

This case illustrates that trials of a variety of noninvasive ventilators may be necessary before an acceptable one is identified. It also shows that many considerations, including pragmatic limitations in the home, help to determine the final selection. Here the rocking bed was selected not because of greater efficacy or patient preference but because of greater ease of application as perceived by the patient's wife.

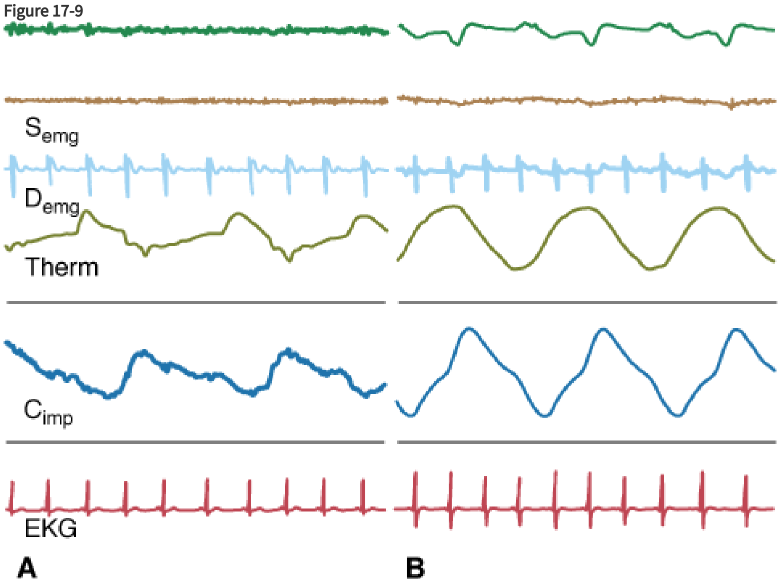
Case 2: Use of the Pneumobelt

M.C., a 24-year-old man with Duchenne muscular dystrophy, presented with a weakening voice, morning headaches, and hypersomnolence. Pulmonary function studies showed severe restriction with reductions of both FVC and FEV_1 to 0.5 L (14% of predicted). Arterial blood-gas determinations on room air showed a pH of 7.36, a Pa_{CO_2} of 68 mm Hg, and a Pa_{O_2} of 55 mm Hg. The patient was admitted to the hospital for a trial of noninvasive ventilation.

Noninvasive positive-pressure ventilation was tried initially, but the patient could not tolerate a face mask or lip seal. (Comfortable nasal masks were not available commercially at that time.) Negative-pressure ventilators were tried, but the patient found the iron lung too bulky and the pneumowrap too restricting and uncomfortable. Standard chest shells fit poorly because of his mild scoliosis. Only the pneumobelt, which was quite effective in augmenting his minute volume, was acceptable to the patient. His spontaneous respiratory rate was 28 breaths/min, tidal volume was 90 mL, and minute volume was 2.5 L/min. The pneumobelt was set at a rate of 22 breaths/min and a positive pressure of 35 cm H_2O, producing a tidal volume of 200 mL and a minute volume of 4.4 L/min. Arterial blood-gas determination after 2 hours during the initial trial showed a pH of 7.41, a Pa_{CO_2} of 59 mm Hg, and a Pa_{O_2} of 78 mm Hg.

The patient was discharged with instructions to use the pneumobelt at night and as needed during the daytime. Arrangements were made to attach a ventilator platform to his wheelchair to facilitate daytime use. The patient rapidly adapted to nocturnal use and encountered little difficulty in sleeping while sitting upright. In addition to nocturnal use, he used the ventilator for 2 to 3 hours during the daytime. A daytime room-air arterial blood-gas determination during spontaneous breathing after full adaptation 7 months after initiation showed a pH of 7.37, a Pa_{CO_2} of 49 mm Hg, and a Pa_{O_2} of 84 mm Hg. A nocturnal polysomnogram showed excellent synchrony of chest wall movement with the ventilator and no evidence of obstructive apneas (Fig. 17-9). Oxygen saturations remained between 95% and 98% throughout the night while the patient was breathing room air.

Figure 17-9



Nocturnal recording of case 2. **A.** Spontaneous breathing at a rate of 20 breaths/min. As expected, airflow increases, whereas chest wall dimensions decrease. **B.** During use of pneumobelt at a rate of 22 breaths/min, chest wall and ventilator are synchronized. However, chest wall motion is paradoxical, expanding during expiration. Diaphragmatic electromyogram (D_{emg}) failed to detect electrical muscular discharge during either spontaneous breathing or ventilator use. C_{imp}, chest wall impedance; *ECG,* electrocardiogram; *EMG,* electromyelogram; *Semg,* submental EMG; *THERM,* nasal thermistor.

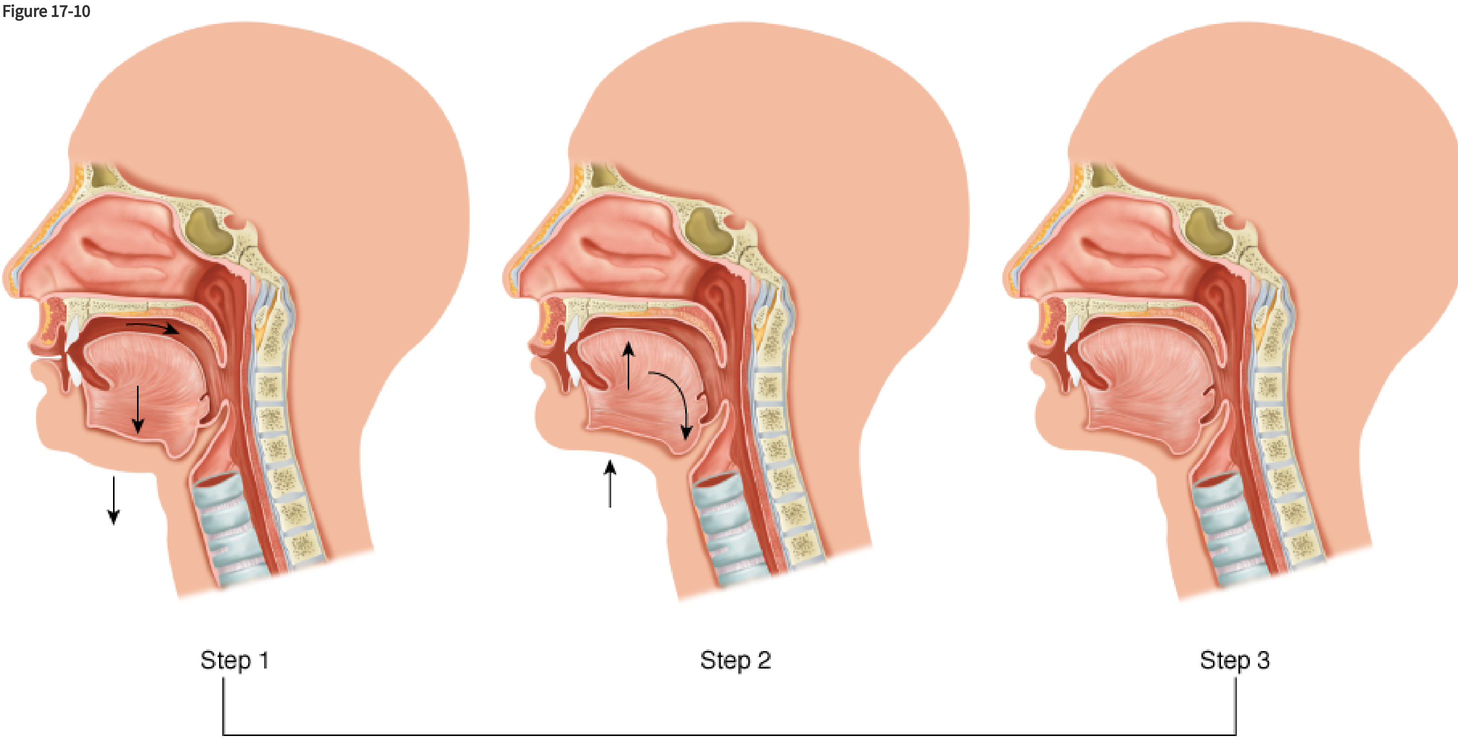
The patient continued to use the pneumobelt both nocturnally and during the daytime for the next 2 years and found it particularly useful for reducing dyspnea associated with eating. During this time, his pulmonary function deteriorated gradually, necessitating increasing daytime use of the pneumobelt. Eventually, he had very little free time from the ventilator, amounting to 1 to 2 hours/day. Shortly, in September 1985, the patient developed pneumonia and required endotracheal intubation because of airway secretions that were unmanageable with either the pneumobelt or an iron lung, despite manual cough assistance (mechanical assistance was not tried). He was unable to resume pneumobelt use, and a tracheostomy was performed. He lived at home for an additional 3 years following tracheostomy placement but died unexpectedly when his ventilator inadvertently became detached from his tracheostomy tube.

This case shows that the pneumobelt may be used as the main mode of ventilator support in exceptional patients who prefer it to other forms of noninvasive ventilation. This patient was able to adapt quickly to nocturnal use while sleeping in the sitting position. The pneumobelt was an appropriate choice for him not only because he preferred it but also because he had a favorable body habitus and slowly progressive respiratory failure that allowed a several-week adaptation period. The pneumobelt proved to be temporizing in this patient, although it provided adequate ventilator assistance even when he was almost entirely dependent on ventilator support. It was inadequate, however, when he developed a pulmonary infection, illustrating the limitations of the device in the face of acute respiratory failure.

Glossopharyngeal Breathing

Glossopharyngeal, or "frog," breathing was first described by Dail et al[28] after they observed a polio patient who had begun using it spontaneously. The technique consists of repetitive gulping motions in which the tongue thrusts small boluses of air into the lungs (Fig. 17-10). The 40-mL to 80-mL boluses take roughly 0.5 second and are repeated eight to twelve times in succession to achieve a tidal volume of 500 to 600 mL, followed by passive exhalation. Roughly ten of these tidal volume maneuvers are performed each minute so that a minute volume of 4 to 8 L/min can be attained. In this way, persons with severely weakened respiratory muscles can use the technique to sustain ventilation and free themselves from the need for continuous ventilator assistance. Patients who become adept at the technique typically use it for periods ranging from a few minutes to several hours, but some ventilator-dependent patients can use it for up to 14 consecutive hours without other aids to ventilation (Sternburg L, personal communication).

Figure 17-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Illustration of steps taken by patient using glossopharyngeal breathing. In *step 1,* the mouth fills with air as the tongue is retracted. In *step 2,* the lips are closed and the tongue is raised to the roof of the mouth forcing air through the open glottis. In *step 3,* the larynx is closed, trapping air in the lungs, and the cycle is then repeated. (Used, with permission, from Maltais F. Glossopharyngeal breathing. *Am J Respir Crit Care Med.* 2011;184:381).

Glossopharyngeal breathing also may be used to aid spontaneous breathing in patients with marginal pulmonary reserve by supplementing each tidal breath with 1 to 2 boluses of air or to assist coughing. Coughing is assisted by repeating boluses until tidal volumes of 2 to 2.5 L/min are achieved, allowing greater expiratory flows.[28,29] Complications are infrequent, with occasional patients complaining of aerophagia or chest tightness when fully inflating their lungs to assist cough.

Glossopharyngeal breathing requires intact function of upper airway structures, including the glottis, and therefore is of little value in patients with severe global weakness or bulbar dysfunction. Efficiency is also impaired by reduced chest wall or lung compliance or increased airway resistance, limiting use in patients with chest wall deformities or chronic obstructive pulmonary disease. In addition, because the upper airway contractions are voluntary, it cannot be used during sleep. Use of the technique complements any of the noninvasive aids to ventilation to extend "free time" from the ventilator.[30]

Glossopharyngeal breathing was used widely during the 1950s when it was taught to many patients with respiratory paralysis secondary to poliomyelitis. It is used less often today, however, partly because skilled teachers are few and also because most conditions leading to chronic respiratory insufficiency are associated with global muscle weakness or alterations in respiratory system compliance or airway resistance. Nonetheless, occasional glossopharyngeal breathers are found among survivors of the polio epidemics,[29] and some patients with muscular dystrophy or cervical spinal cord lesions learn to use the technique, sometimes spontaneously. Illustrative is a recent case study of a 6-year-old boy with a high cervical lesion secondary to tumor resection who learned glossopharyngeal breathing on his own and was able to use it for ventilator-free breathing for up to 12 hours a day during a 16-year period. Despite having a tracheostomy, he was able to achieve a vital capacity of 3.3 L to augment cough.[30]

References

1. Eve FC. Actuation of the inert diaphragm. *Lancet.* 1932;2:995–997.
2. Eve FC. Resuscitation methods for rescue boats. *Br Med J.* 1945;1:21–22. [PubMed: 20785834]
3. Gordon AS, Fainer DC, Ivy AC. Artificial respiration: a new method and a comparative study of different methods in adults. *JAMA.* 1950;144:1455–1465. [PubMed: 14794366]
4. Gordon AS, Raymon F, Savode M, Ivy AC. Manual of artificial respiration: comparison of effectiveness of various methods on apneic normal adults. *JAMA.* 1950;144:1447–1452. [PubMed: 14794363]
5. Wright J. The Respiraid rocking bed in poliomyelitis. *Am J Nurs.* 1947;47:454–455. [PubMed: 20254536]
6. Carter HA. McKesson Respiraid rocking bed accepted. *JAMA.* 1950;144:1181.
7. Lewis L, Hirschberg GG, Adamson JP. Respiratory rehabilitation in poliomyelitis. *Arch Phys Med Rehabil.* 1957;38:243–249. [PubMed: 13412276]
8. Sternburg L, Sternburg D. *View from the Seesaw.* New York, NY: Dodd, Mead, 1986.
9. Adamson JP, Lewis L, Stein JD. Application of abdominal pressure for artificial respiration. *JAMA.* 1959;169:1613–1617. [PubMed: 13640911]
10. Janelli HV, Krouskop TA, Canzoneri J, Jameson J. Positive pressure respiratory assist for wheelchair-mobile persons. *Arch Phys Med Rehabil.* 1980;61:143–144. [PubMed: 6989344]
11. Gray FD, Field AS. The use of mechanical assistance in treating cardiopulmonary disease. *Am J Med Sci.* 1959;288:146–152.
12. Plum F, Whedon GD. The rapid-rocking bed: its effect on the ventilation of poliomyelitis patients with respiratory paralysis. *N Engl J Med.* 1951;245:235–241. [PubMed: 14863562]
13. Colville P, Shugg C, Ferris BG. Effects of body tilting on respiratory mechanics. *J Appl Physiol.* 1956;9:19–24. [PubMed: 13357407]
14. Joos TH, Dickinson DG, Talner NS, Wilson JL. The rocking bed and head position. *N Engl J Med.* 1956;255:1089–1090. [PubMed: 13378619]
15. Yang G-FW, Alba A, Lee M, Khan A. Pneumobelt for sleep in the ventilatory user: clinical experience. *Arch Phys Med Rehabil.* 1989;70:707–711. [PubMed: 2774892]
16. Miller HJ, Thomas E, Wilmot CB. Pneumobelt use among high quadriplegic population. *Arch Phys Med Rehabil.* 1988;69:369–372. [PubMed: 3163246]

17. Hill NS. Clinical application of body ventilators. *Chest.* 1986;90:897–905. [PubMed: 3536343]

18. Mohr CH, Hill NS. Long-term follow-up of nocturnal ventilatory assistance in patients with respiratory failure due to Duchenne-type muscular dystrophy. *Chest.* 1990;97:91–96. [PubMed: 2104794]

19. Hill NS, Redline S, Carskadon MA, et al. Sleep-disordered breathing in patients with Duchenne muscular dystrophy using negative pressure ventilators. *Chest.* 1992;102:1656–1662. [PubMed: 1446467]

20. Abd AG, Braun NMT, Baskin MI, et al. Diaphragmatic dysfunction after open heart surgery: treatment with a rocking bed. *Ann Intern Med.* 1989;111:881–886. [PubMed: 2573307]

21. Chalmers RM, Howard RS, Wiles CM, Spencer GT. Use of the rocking bed in the treatment of neurogenic respiratory insufficiency. *Q J Med.* 1994;87:423–429. [PubMed: 7922294]

22. Bach JR, Alba AS. Total ventilatory support by the intermittent abdominal pressure ventilator. *Chest.* 1991;99:630–636. [PubMed: 1899821]

23. Ellis ER, Bye PT, Bruderer JW, Sullivan CE. Treatment of respiratory failure during sleep in patients with neuromuscular disease: positive-pressure ventilation through a nose mask. *Am Rev Respir Dis.* 1987;135:148–152. [PubMed: 3541713]

24. Cormican LJ, Higgins S, Davidson AC, et al. Rocking bed and prolonged independence from nocturnal non-invasive ventilation in neurogenic respiratory failure associated with limb weakness. *Postgrad Med J.* 2004;80:360–362. [PubMed: 15192173]

25. Bryce-Smith R, Davis HS. Tidal exchange in respirators. *Curr Res Anesth Analg.* 1954;33:73–85. [PubMed: 13141699]

26. Goldstein RS, Molotiu N, Skrastins R, et al. Assisting ventilation in respiratory failure by negative pressure ventilation and by rocking bed. *Chest.* 1987;92:470–474. [PubMed: 3476256]

27. Goldberg AI, Cane RD, Childress D, et al. Combined nasal intermittent positive-pressure ventilation and rocking bed in chronic respiratory insufficiency: nocturnal ventilatory support of a disabled person at home. *Chest.* 1991;99:627–629. [PubMed: 1899820]

28. Dail CW, Affeldt JE, Collier CR. Clinical aspects of glossopharyngeal breathing: report of use by one hundred postpoliomyelitis patients. *JAMA.* 1953;158:445–449.

29. Bach JR, Alba AS, Bodofsky E, et al. Glossopharyngeal breathing and noninvasive aids in the management of post-polio respiratory insufficiency. *Birth Defects Orig Artic Ser.* 1987;23:99–113. [PubMed: 3304449]

30. Bianchi C, Grandi M, Felisari G. Efficacy of glossopharyngeal breathing for a ventilator-dependent, high-level tetraplegic patient after cervical cord tumor resection and tracheotomy. *Am J Phys Med Rehabil.* 2004;83:216–219. [PubMed: 15043357]